Data Engineering Pipeline: Implementation Plan

This document provides a detailed breakdown of all features implemented in this project, explaining the technology stack, minute technical details, and how the system works in a simple way.

□□ 1. Technology Stack

The project leverages a robust stack designed for professional data engineering:

- **PySpark**: The core data processing engine for handling large datasets.
  - *Simple Way:* The "muscle" that does the heavy work of moving and cleaning data.
- **Delta Lake (Delta-Spark)**: Adds reliability and transactional capabilities to Spark.
  - *Simple Way:* A "safety net" that prevents data loss and allows you to "time travel" to older versions of data.
- **Streamlit**: The framework used to build the interactive dashboard.
  - *Simple Way:* The "face" of the project where you see your data and charts.
- **Pandas & Delta Table (Python)**: Used for fast, in-memory reading of data for the dashboard.
  - *Simple Way:* The "speed lane" that makes the dashboard feel fast and responsive.
- **WinUtils**: A set of helper tools to make Hadoop/Spark work seamlessly on Windows.
  - *Simple Way:* The "translator" that helps Spark understand Windows folders.

□□ 2. Infrastructure & Windows Optimization

Running a big-data engine like Spark on Windows requires specific "under-the-hood" work:

- **WinUtils Automation**: A script that automatically downloads necessary components (`winutils.exe`) and sets up environment variables.
- **JVM Module Unlocks**: Configuration of 14 specific security flags (`--add-opens`) to allow Spark to access system memory on modern Java versions (17+).
- **Permissions Shimming**: Telling Spark to ignore Windows security restrictions when saving data files to prevent "Access Denied" errors.
- **Native Sync**: Ensuring the dashboard and the processing engine use the exact same version of Python.

□ 3. Minute Features Implementation

□ Bronze Layer: Raw Ingestion

- **Smart Header Detection**: A keyword-based scanner that identifies column headers even if they are renamed or slightly different.
- **Dynamic Extension Routing**: Automatically moves unsupported file types (non-CSV or non-JSON) to a "junk" directory to avoid errors.
- **Hybrid Preview**: Shows data in the dashboard the moment it is saved, before the heavy engine even starts.

□ Silver Layer: The Validation Gate

- **Schema Enforcement**: Strictly validates that data types are correct (e.g., "Revenue" must be a number, not text).

- **The Quarantine System**: Instead of deleting bad data, records with missing IDs are saved to a separate "Quarantine" table for inspection.
- **Clone Killer (Deduplication)**: Automatically identifies duplicate entries and only keeps the single newest version.

Gold Layer: Business Intelligence

- **Daily Totals**: Aggregates all transactions by date so you can see trends over time.
- **ID Aggregation**: Groups many individual order IDs into a single, clean list to keep reports organized.

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4. Performance & Partitioning Details

We applied several advanced techniques to make the system run ~500% faster on a local machine:

- **Shuffle Lane Optimization**: We reduced the number of internal data lanes from 200 down to **1** to eliminate overhead.
  - *Simple Way:* Like turning a 200-lane highway into a 1-lane road when there's only one car—it's much faster to manage.
- **Writer Coalescing**: Ensuring every layer saves its data into **one** single file instead of many tiny fragments.
  - *Simple Way:* Like putting all your papers in one folder instead of scattering 100 pages across the floor.
- **Backslash Translation**: Automatically converting Windows-style paths (`\`) into a format Spark understands (`/`).

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5. Advanced Dashboard Features

- **Edge-Run Animations**: Glowing lines that show where data is flowing in real-time.
- **Rejection Diagnosis**: A system where you can click on a rejected file to see a "Doctor's Report" explaining exactly what was wrong.
- **Delta ACID History**: A searchable "time machine" table that shows every version of your data and who modified it.

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6. How it Works (Simple Workflow)

1. **Drop Data**: Put a file in the `data/input/` folder.
2. **Instant Preview**: The dashboard shows it immediately in the **Bronze** section.
3. **Process**: The engine runs its checks. If data is bad, it goes to the **Rejection Log**.
4. **Clean & Aggregate**: Good data is refined into the **Silver** layer and then summarized into your **Gold** analytics charts automatically.

> [!IMPORTANT] This pipeline is designed to be **Zero-Friction**. It handles the technical complexity of Spark and Windows automatically so you can focus on data insights.